Statistics for Linguists
Lecture 6: More than two groups, more than one outcome

Anton Buzanov Maxim Bazhukov

MA Theoretical Linguistics
May 13, 2026

Today: from two groups to many groups

Previously, when we worked with **numeric outcomes**, the basic question was simple:

Do two groups differ?

Examples:

- Do dialect A and dialect B differ in vowel duration?
- Do grammatical and ungrammatical sentences differ in reaction time?
- Do two constructions differ in acceptability score?

But linguistic data often have **more than two groups**: North vs Central vs South

How do we compare more than two groups safely?

Why this was easier with categorical data

Last time, with categorical data, more than two groups were not a big conceptual problem.

	SOV	SVO	VSO
Dialect A	40	20	10
Dialect B	25	35	10
Dialect C	15	25	30

We could put everything into one contingency table and ask: **Is word order associated with dialect?**

The chi-square test handled the whole table at once. But for numeric data, the temptation is different: **just compare every pair of groups**
That temptation is dangerous.

Plan for 80 minutes

- more than two groups
- why many *t*-tests are dangerous
- family-wise error
- Bonferroni correction
- Holm-Bonferroni correction
- ANOVA as an omnibus test
- between-group variation
- within-group variation
- the *F*-statistic
- post-hoc testing
- Kruskal-Wallis
- rank logic
- MANOVA
- several related outcomes
- linguistic interpretation
- seminar workflow

The tempting but dangerous solution

Suppose we compare F1 of /a/ in three dialects:

North, Central, South

The tempting solution is to run three *t*-tests:

North vs Central

North vs South

Central vs South

But now we are no longer asking one question.

We are asking **three separate questions**.

And every test has a chance of giving us a false positive.

The number of pairwise tests grows quickly

For k groups, the number of pairwise comparisons is:

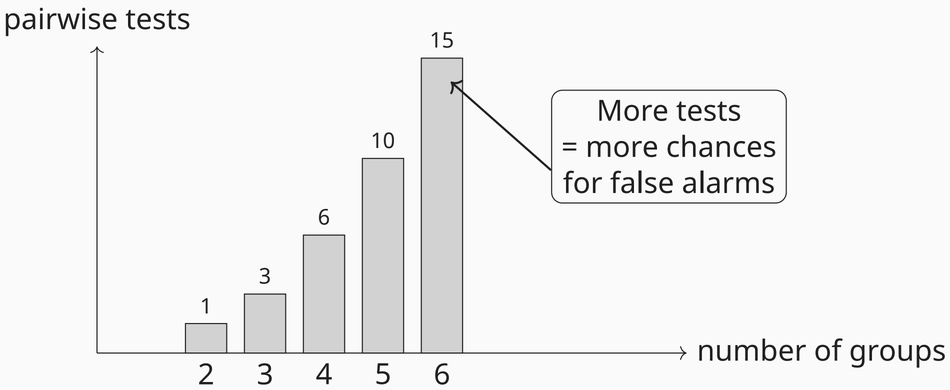
$$\frac{k(k-1)}{2}$$

Groups	Example	Pairwise tests
2	A, B	1
3	A, B, C	3
4	A, B, C, D	6
5	A, B, C, D, E	10
6	A, B, C, D, E, F	15

So with many groups, “just do pairwise tests” quickly becomes:

many chances to fool ourselves

The multiple-testing monster



This is not a technical detail. It changes what we are allowed to claim.

Family-wise error: the core problem

A single test at $\alpha = .05$ means:

5% chance of a false positive if H_0 is true

But if we run many tests, we care about:

the chance of at least one false positive

This is called the **family-wise error rate**.

If we run m independent tests:

$$P(\text{at least one false positive}) = 1 - (1 - .05)^m$$

The family is the whole set of comparisons we decided to inspect.

False positives accumulate

Groups	Pairwise tests	Chance of at least one false positive
2	1	5%
3	3	14%
4	6	26%
5	10	40%
6	15	54%

So if we compare six dialects pairwise, even when there is no real difference, there is about a **54%** chance that at least one comparison looks significant.

That is not discovery.

That is statistical fishing.

Toy example: the fake dialect discovery

Imagine six dialects. In reality, they all have the same mean vowel duration:

$$H_0 : \quad \mu_A = \mu_B = \mu_C = \mu_D = \mu_E = \mu_F$$

But we run all 15 pairwise *t*-tests.

	A	B	C	D	E	F
A	–	.62	.44	.031	.77	.19
B		–	.58	.21	.049	.83
C			–	.67	.34	.12
D				–	.28	.55
E					–	.71
F						–

Two comparisons look “significant” at .05.
But this can easily happen by chance when we test many pairs.

Toy example: before and after correction

Same fake dialect example:

Comparison	Raw *p*	Significant at .05?	After correction?
A–D	.031	yes	no
B–E	.049	yes	no
C–F	.120	no	no
A–B	.620	no	no
D–F	.550	no	no

Correction protects us from treating random noise as a linguistic pattern.

Without correction, we might write:

"Dialect A differs from dialect D."

But that would be overconfident.

Bonferroni correction

Bonferroni is the simplest way to control family-wise error.

$$\alpha_{\text{corrected}} = \frac{\alpha}{m}$$

Usual threshold	Number of tests	Bonferroni threshold
$\alpha = .05$	$m = 10$	$.05/10 = .005$

So after 10 comparisons:

$p < .005$ counts as significant, not merely $p < .05$

Easy and safe, but often conservative.

Bonferroni toy example

Suppose we compare five constructions. That gives:

$$m = \frac{5(5-1)}{2} = 10$$

Bonferroni threshold: $.05/10 = .005$

Comparison	Raw p	Bonferroni result
A–B	.041	not significant
A–C	.018	not significant
A–D	.006	not significant
A–E	.003	significant
B–C	.270	not significant

Only $p = .003$ survives because the corrected threshold is very strict.

Holm-Bonferroni correction

Holm-Bonferroni is also designed to control family-wise error.

But it is less conservative than simple Bonferroni.

Procedure:

1. Sort the p-values from smallest to largest.
2. Compare the smallest p to α/m.
3. Compare the next p to $\alpha/(m-1)$.
4. Continue with gradually less strict thresholds.
5. Stop when a comparison fails.

Idea:

the strongest effects face the strictest test first

Holm-Bonferroni toy example

Suppose we have five pairwise tests.

Sort the *p*-values:

Order	Raw *p*	Holm threshold	Pass?	Decision
1	.006	.010	yes	keep going
2	.018	.0125	no	stop
3	.021	.0167	–	not significant
4	.140	.025	–	not significant
5	.310	.050	–	not significant

Only the smallest *p*-value survives.

Holm is usually preferable to plain Bonferroni when you want a simple family-wise correction.

Why correction matters linguistically

Without correction, a paper may accidentally turn noise into a story.

But if these claims come from many uncorrected comparisons, they may be fragile.

Correction asks:

Would this result still look convincing

after we admit how many comparisons we tried?

So what is ANOVA for?

ANOVA helps us avoid starting with many pairwise tests.

Instead of asking:

A vs B?   A vs C?   B vs C?

we first ask one omnibus question:

$$H_0: \quad \mu_A = \mu_B = \mu_C$$

That is:

Is there evidence that not all group means are equal?

Only after that do we inspect pairwise differences with correction.

Today's running example

A phonetician compares vowel realisation in three dialects.

Dialect	North	Central	South
Mean F1 of /a/	710 Hz	760 Hz	830 Hz

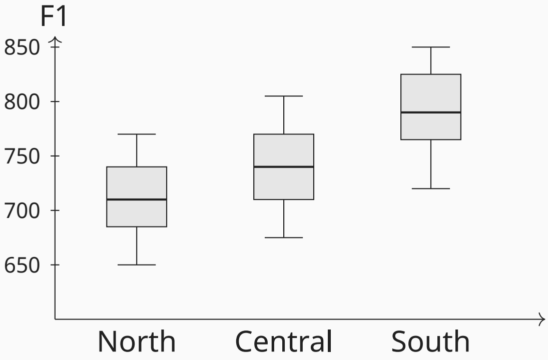
Question:

Do the dialects differ in vowel height?

But there are three groups, not two.

North vs Central, North vs South, Central vs South

First look: three dialects



The groups look different, but we need a model of uncertainty.

ANOVA: the big idea

ANOVA means:

Analysis of Variance

But the usual question is about means:

$$H_0 : \mu_1 = \mu_2 = \mu_3 = \cdots = \mu_k$$

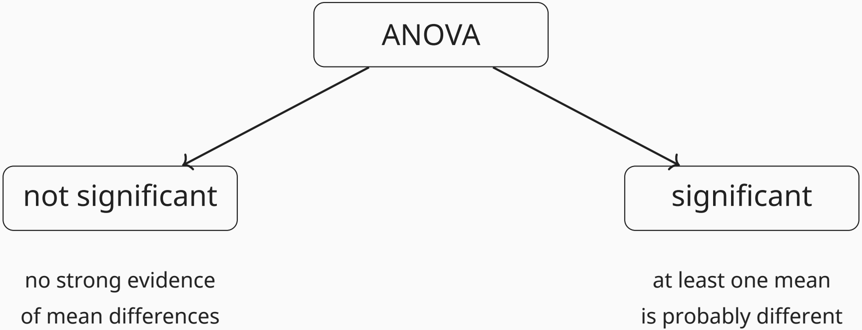
For the dialect example:

$$H_0 : \mu_{\text{North}} = \mu_{\text{Central}} = \mu_{\text{South}}$$

ANOVA asks one omnibus question:

Is there evidence that at least one group mean differs?

Omnibus means: one big alarm bell



ANOVA does not immediately tell us which groups differ.

What ANOVA compares

ANOVA compares two kinds of variability.

Between-group variation

How far are the group means from the grand mean?

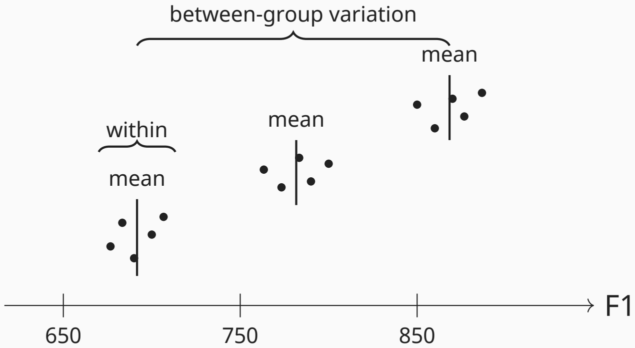
Within-group variation

How scattered are observations inside each group?

$$F = \frac{\text{between-group variation}}{\text{within-group variation}}$$

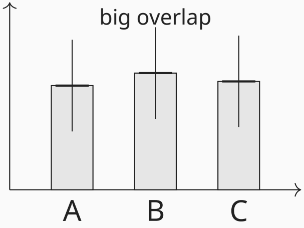
Large F: groups are far apart relative to internal noise.

Between vs within: picture

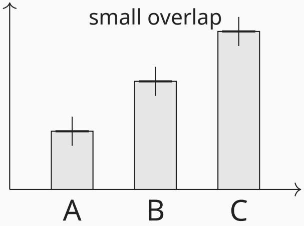


Two possible worlds

World A: weak evidence



World B: strong evidence



ANOVA is impressed by large mean differences and small within-group noise.

ANOVA vocabulary

Term	Meaning
Factor	categorical predictor, e.g. dialect
Level	group inside a factor, e.g. North, Central, South
Dependent variable	numeric outcome, e.g. F1
Omnibus test	one global test for any group difference
Post-hoc test	follow-up pairwise comparison
Residual	observation minus model prediction

In a one-way ANOVA, there is one categorical predictor and one numeric outcome.

One-way ANOVA model

For observation j in group i:

$$Y_{ij} = \mu + \alpha_i + \varepsilon_{ij}$$

Where:

- Y_{ij} is the observed value, for example F1;
- μ is the grand mean;
- α_i is the effect of group i;
- ε_{ij} is residual noise.

Null hypothesis:

$$\alpha_1 = \alpha_2 = \cdots = \alpha_k = 0$$

The ANOVA table

Source	Sum Sq	df	Mean Sq	F
Group	SS_B	$k-1$	MS_B	MS_B/MS_W
Residual	SS_W	$N-k$	MS_W	
Total	SS_T	$N-1$		

Important idea:

$$SS_T = SS_B + SS_W$$

Total variability is split into explained and unexplained variability.

A tiny ANOVA example

Dialect	Observation 1	Observation 2	Mean
North	700	720	710
Central	750	770	760
South	820	840	830

Grand mean:

$$\bar{Y} = 766.7$$

The group means are not identical:

$$710, \quad 760, \quad 830$$

ANOVA asks whether this separation is large relative to the within-group scatter.

Omnibus result: what it means

Suppose ANOVA gives:

$$F(2, 57) = 18.4, \quad p < .001$$

This means:

The three dialect means are unlikely to be all equal.

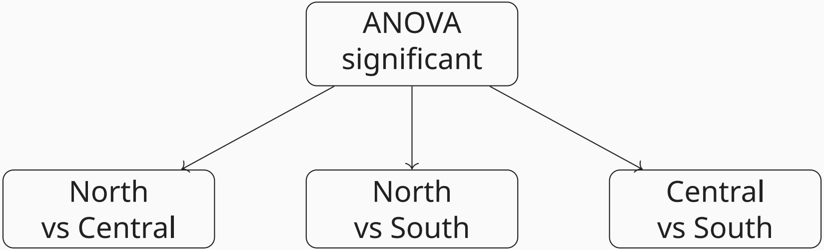
It does **not** mean: North differs from Central or Central differs from South

For that, we need post-hoc comparisons.

After ANOVA: who differs from whom?

ANOVA is the doorbell.

Post-hoc tests are opening the rooms.



pairwise tests must be corrected

Tukey HSD

A common post-hoc method after ANOVA is:

Tukey's Honest Significant Difference

It compares all pairs of group means while controlling family-wise error.

Example output:

Comparison	Mean difference	Adjusted p
North - Central	-50	.041
North - South	-120	$< .001$
Central - South	-70	.008

Now we can say which dialect pairs differ.

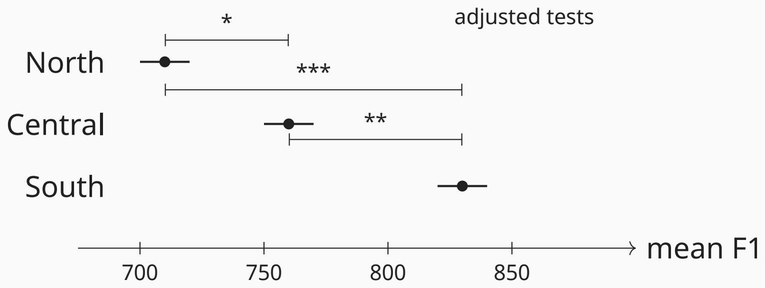
Correction methods: quick overview

Method	Typical use
Bonferroni	simple, conservative family-wise correction
Holm-Bonferroni	simple, less conservative family-wise correction
Tukey HSD	all pairwise group mean comparisons after ANOVA
Benjamini-Hochberg	controls false discovery rate, useful with many exploratory tests

The slogan:

Do not let one noisy comparison become a discovery.

Visualising post-hoc results



ANOVA assumptions

Classic one-way ANOVA assumes:

1. independent observations;
2. approximately normal residuals;
3. similar variance across groups.

For linguists, the first one is often the most dangerous.

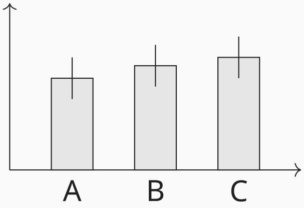
Repeated measurements from the same speaker or item are not independent:

$$\text{speaker 1 says 50 vowels} \neq \text{50 independent speakers}$$

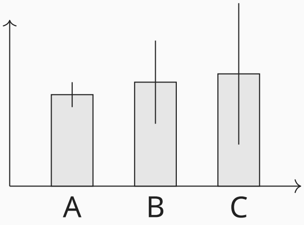
That problem will lead us to mixed models later in the course.

Assumption check: equal variances

Looks okay

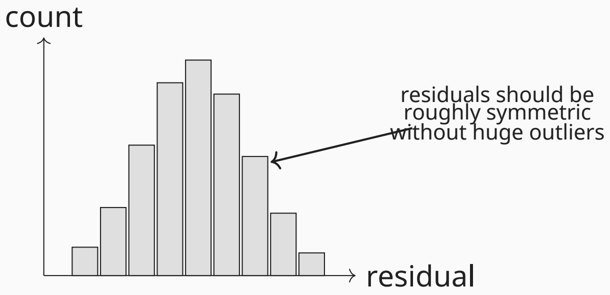


Looks suspicious



If variances are very different, consider Welch ANOVA or a more robust model.

Assumption check: residuals



ANOVA is fairly robust, but terrible residuals should make us pause.

Effect size: eta-squared

A tiny *p*-value does not say how large the effect is.

One ANOVA effect size:

$$\eta^2 = \frac{SS_B}{SS_T}$$

Interpretation:

proportion of total variation explained by group

Example:

$$\eta^2 = .32$$

About 32% of the variation in F1 is associated with dialect.

Reporting ANOVA

A useful report includes:

- the outcome and grouping variable;
- the omnibus ANOVA result;
- effect size;
- post-hoc comparisons with correction;
- linguistic interpretation.

Example:

$$F(2, 57) = 18.4, \quad p < .001, \quad \eta^2 = .39$$

Southern speakers had the highest F1, suggesting a lower realisation of /a/.

When ANOVA feels too optimistic

Sometimes the outcome is numeric, but ANOVA is not comfortable.

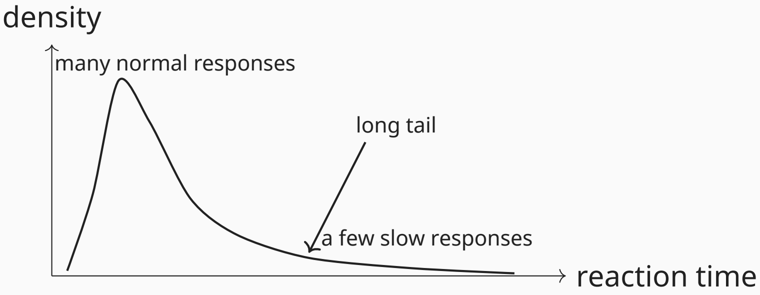
Examples:

- reaction times with a long right tail;
- acceptability ratings on a 1–7 scale;
- very small samples;
- many outliers;
- ordinal judgement data.

Then we can use a rank-based alternative:

Kruskal-Wallis test

Skewed data: reaction times



The mean can be pulled around by a small number of slow trials.

Kruskal-Wallis: the big idea

Kruskal-Wallis does not compare raw means.

It compares **ranks**.

$$\text{smallest value} = 1, \quad \text{next smallest} = 2, \quad \ldots$$

If groups come from the same distribution, their ranks should be mixed.

If one group tends to have high values, it will also have high ranks.

Rank logic with linguistic ratings

Item	A	B	C	D	E	F	G	H	I
Condition	Base.	Base.	Base.	Focus	Focus	Focus	Topic	Topic	Topic
Rating	2	3	4	4	5	5	6	6	7
Rank	1	2	3.5	3.5	5.5	5.5	7.5	7.5	9

Topic items tend to receive high ranks.

Baseline items tend to receive low ranks.

That is the kind of pattern Kruskal-Wallis detects.

Kruskal-Wallis hypotheses

Informally:

$$H_0: \quad \text{the groups have the same distribution}$$

Alternative:

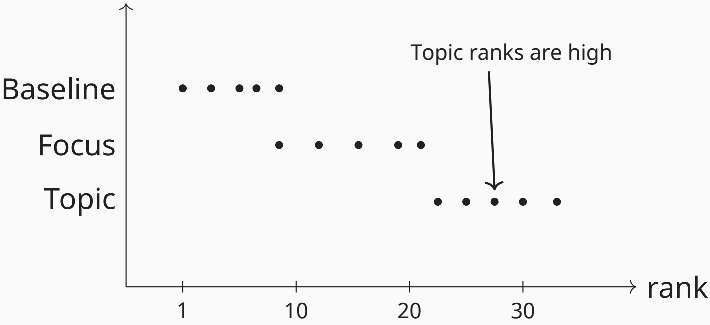
at least one group tends to have different values

Important:

Kruskal-Wallis is an omnibus test too.

It tells us that some difference exists, not exactly where it is.

Kruskal-Wallis visual intuition



After Kruskal-Wallis

If Kruskal-Wallis is significant, we still need post-hoc tests.

Common choices: Dunn's test or pairwise rank-based tests with correction.

Again:

post-hoc tests must be corrected for multiple comparisons.

Example:

Comparison	Difference in mean ranks	Adjusted p
Baseline - Focus	-5.2	.090
Baseline - Topic	-13.6	.002
Focus - Topic	-8.4	.030

ANOVA vs Kruskal-Wallis

Question	ANOVA	Kruskal-Wallis
Outcome	numeric	numeric or ordinal
Main comparison	means	ranks/distributions
Robust to outliers	less	more
Assumptions	stronger	weaker
Post-hoc needed?	yes	yes
Typical linguistic use	acoustic means	ratings, skewed RTs

Kruskal-Wallis is not “ANOVA but magic”.
It changes the question from means to rank patterns.

What if there is more than one outcome?

Linguistic measurements often come in families.

For vowels:

$$F1, \quad F2, \quad \text{duration}, \quad \text{intensity}$$

For experimental responses:

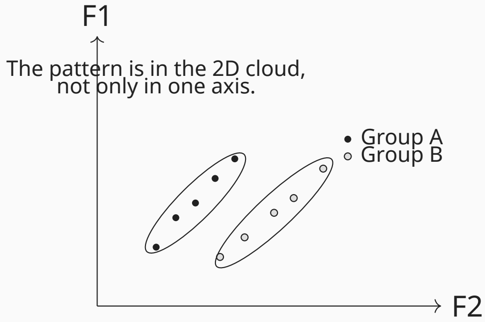
$$\text{accuracy}, \quad \text{reaction time}, \quad \text{confidence}$$

For texts:

$$\text{mean sentence length}, \quad \text{lexical diversity}, \quad \text{embedding depth}$$

Do groups differ in a multidimensional outcome space?

Separate ANOVAs can miss the shape



MANOVA tests group differences across several related dependent variables at once.

MANOVA: the big idea

MANOVA means:

Multivariate Analysis of Variance

Instead of one outcome:

$$Y$$

we have a vector of outcomes:

$$\mathbf{Y} = \begin{bmatrix} F1 \\ F2 \\ duration \\ intensity \end{bmatrix}$$

MANOVA asks: **Do the groups differ in their multivariate mean vectors?**

MANOVA hypotheses

For three dialects and four acoustic measures:

$$H_0 : \boldsymbol{\mu}_{North} = \boldsymbol{\mu}_{Central} = \boldsymbol{\mu}_{South}$$

where each mean is a vector:

$$\boldsymbol{\mu}_{North} = \begin{bmatrix} \mu_{F1} \\ \mu_{F2} \\ \mu_{duration} \\ \mu_{intensity} \end{bmatrix}$$

The test is about the whole acoustic profile, not one measure alone.

A linguistic MANOVA dataset

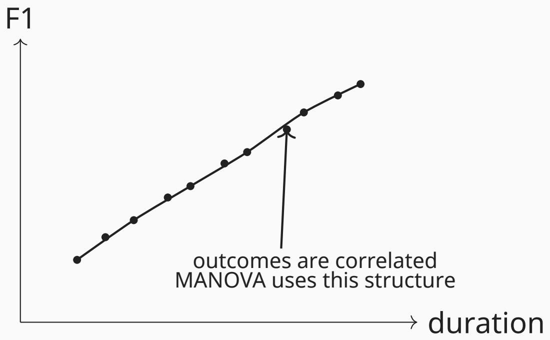
Token	Dialect	F1	F2	Duration
1	North	710	1320	92
2	North	690	1350	88
3	Central	760	1260	101
4	Central	775	1240	104
5	South	830	1180	119
6	South	845	1160	122

Here the dependent variables are related:

$$F1, \quad F2, \quad duration$$

So MANOVA is more natural than three isolated ANOVAs.

Why related outcomes matter



If outcomes are strongly related, analysing them jointly can be more informative.

MANOVA test statistics

MANOVA output often gives several statistics:

Statistic	Informal idea
Wilks’ Lambda	unexplained multivariate variation
Pillai’s trace	robust overall separation
Hotelling-Lawley trace	another separation measure
Roy’s greatest root	strongest single dimension

In practice, many people prefer Pillai’s trace because it is relatively robust.

But the main interpretation is still:

Do groups differ in the combined outcome space?

Statsmodels MANOVA

In Python, MANOVA is available in `statsmodels`.

The formula can look like this:

```
F1 + F2 + duration + intensity ~ dialect
```

Important detail:

statsmodels implements MANOVA through multivariate regression.

So predictors do not have to be only categorical.

For example:

```
F1 + F2 ~ dialect + speech_rate
```

This is useful for linguists because covariates are everywhere.

After MANOVA: what next?

A significant MANOVA says:

groups differ somewhere in the multivariate outcome space

Then inspect:

- plots of the outcome space;
- univariate ANOVAs for each dependent variable;
- corrected post-hoc comparisons;
- effect sizes;
- the linguistic meaning of the combined profile.

MANOVA is the map.
Follow-up analysis tells us which roads matter.

MANOVA assumptions

MANOVA has stronger assumptions than one-way ANOVA.

Common concerns:

- independent observations;
- multivariate normality within groups;
- similar covariance matrices across groups;
- enough observations for the number of outcomes.

Practical warning:

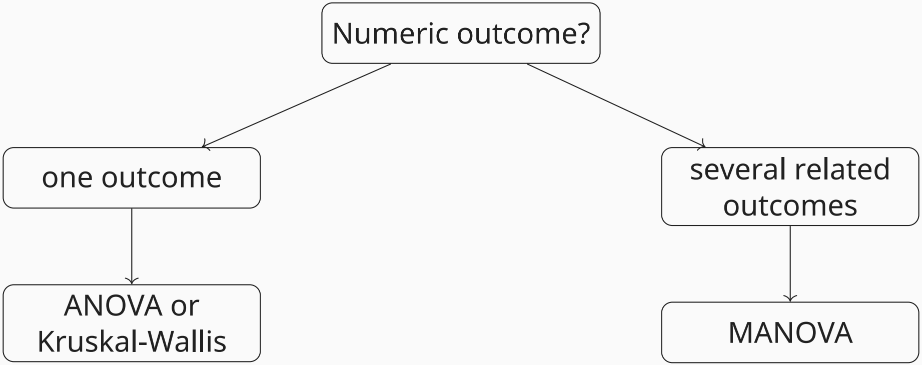
Do not put every available measurement into MANOVA.

Use related variables motivated by a linguistic question.

The key decision

Ask two questions:

How many groups? **How many related outcomes?**



Decision table

Data situation	Linguistic example	Method
3+ groups, one numeric outcome	F1 across dialects	ANOVA
3+ groups, ordinal/skewed outcome	acceptability ratings across conditions	Kruskal-Wallis
3+ groups, several related outcomes	F1, F2, duration across dialects	MANOVA
Many speakers/items repeated	tokens nested in speaker and word	mixed models later

Mini linguistic zoo

Question	Outcome(s)	Method
Do three dialects differ in vowel duration?	duration	ANOVA
Do four constructions differ in acceptability?	1–7 rating	Kruskal-Wallis
Do learner groups differ in RT?	reaction time	ANOVA or Kruskal-Wallis
Do dialects differ in vowel quality?	F1 and F2	MANOVA
Do registers differ in complexity profile?	length, depth, diversity	MANOVA

The method follows the structure of the question.

What you should remember

- Numeric data with more than two groups need special care.
- Many pairwise tests inflate the family-wise error rate.
- Bonferroni and Holm-Bonferroni protect against false positives.
- ANOVA extends mean comparison to more than two groups.
- ANOVA is an omnibus test: it says that some difference exists.
- Post-hoc tests tell us which groups differ.
- Kruskal-Wallis is the rank-based alternative for ordinal or messy data.
- MANOVA compares groups across several related outcomes at once.

Final 60-second summary

With two numeric groups, we could compare two means.

With more than two groups, we must avoid uncontrolled pairwise testing.

So:

one numeric outcome, 3+ groups $\Rightarrow$ ANOVA or Kruskal-Wallis

And:

several related numeric outcomes $\Rightarrow$ MANOVA

The big linguistic question:

Are groups different in the way grammar, sound, or processing is realised?